

99.500 Applied Mathematics for Engineering

Homework 5 Solution

April 7, 2026

1. From Homework 1 Question 1, the non-steady-state temperature equation, BCs and IC for $T(x, t)$ is

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2} \quad (1)$$

$$\text{IC : } T(x, 0) = T_i \quad (2)$$

$$\text{BC1 : } T(0, t) = T_i + \alpha t \quad (3)$$

$$\text{BC2 : } T(x \rightarrow \infty, t) = T_i \quad (4)$$

where

$$\kappa = \frac{k}{\rho c} \quad (5)$$

Take the Laplace transform of Equation (1) to have

$$s\bar{T} - T(x, 0) = \kappa \frac{d^2 \bar{T}}{dx^2} \implies \frac{d^2 \bar{T}}{dx^2} - \frac{s}{\kappa} \bar{T} = -\frac{T_i}{\kappa} \quad (6)$$

The general solution for Equation (6) is

$$\bar{T} = A_1 e^{\sqrt{\frac{s}{\kappa}} x} + A_2 e^{-\sqrt{\frac{s}{\kappa}} x} + \frac{T_i}{s} \quad (7)$$

The Laplace transformed BCs are

$$\text{BC1 : } \bar{T}(0) = \frac{T_i}{s} + \frac{\alpha}{s^2} \implies A_1 = 0 \quad (8)$$

$$\text{BC2 : } \bar{T}(x \rightarrow \infty) = \frac{T_i}{s} \implies A_2 = \frac{\alpha}{s^2} \quad (9)$$

Hence,

$$\bar{T} = \frac{\alpha}{s^2} e^{-\sqrt{\frac{s}{\kappa}} x} + \frac{T_i}{s} \quad (10)$$

The heat flux at the surface of the slab is

$$q = -k \left. \frac{\partial T}{\partial x} \right|_{x=0} \quad (11)$$

$$\implies \mathcal{L}(q) = -k \left. \frac{\partial \bar{T}}{\partial x} \right|_{x=0} = k \frac{\alpha}{s^2} \sqrt{\frac{s}{\kappa}} = \frac{k\alpha}{\sqrt{s^3 \kappa}} \quad (12)$$

Take the inverse Laplace transform of Equation (12) to obtain

$$q = \frac{k\alpha}{\sqrt{\kappa}} 2\sqrt{\frac{t}{\pi}} = 2k\alpha\sqrt{\frac{t}{\kappa\pi}} \quad (13)$$

2. Take the Laplace transform of

$$\frac{\partial T}{\partial t} = \beta \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) \quad (14)$$

to obtain

$$s\bar{T} - T_i = \beta \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \bar{T}}{\partial r} \right) \implies r^2 \frac{d^2 \bar{T}}{dr^2} + r \frac{d\bar{T}}{dr} - \frac{s}{\beta} r^2 \bar{T} = -r^2 \frac{T_i}{\beta} \quad (15)$$

Equation (15) is a modified Bessel differential equation with the homogeneous and particular solutions,

$$\bar{T}_h = A_1 I_0 \left(\sqrt{\frac{s}{\beta}} r \right) + A_2 K_0 \left(\sqrt{\frac{s}{\beta}} r \right) \quad (16)$$

$$\bar{T}_p = \frac{T_i}{s} \quad (17)$$

Hence,

$$\bar{T}(r) = A_1 I_0 \left(\sqrt{\frac{s}{\beta}} r \right) + A_2 K_0 \left(\sqrt{\frac{s}{\beta}} r \right) + \frac{T_i}{s} \quad (18)$$

For the BCs,

$$\text{BC1: } \frac{\partial T}{\partial r} \Big|_{r=0} = 0 \implies \frac{d\bar{T}}{dr} \Big|_{r=0} = 0 \implies A_2 = 0 \quad (19)$$

$$\text{BC2: } T(r_0, t) = T_0 \implies \bar{T}(r_0) = \frac{T_0}{s} \implies A_1 = \frac{T_0 - T_i}{s I_0 \left(\sqrt{\frac{s}{\beta}} r_0 \right)} \quad (20)$$

Hence,

$$\bar{T}(r) = (T_0 - T_i) \frac{I_0 \left(\sqrt{\frac{s}{\beta}} r \right)}{s I_0 \left(\sqrt{\frac{s}{\beta}} r_0 \right)} + \frac{T_i}{s} \quad (21)$$

$$T(r, t) = \mathcal{L}^{-1} \{T\} = (T_0 - T_i) \mathcal{L}^{-1} \left\{ \frac{I_0 \left(\sqrt{\frac{s}{\beta}} r \right)}{s I_0 \left(\sqrt{\frac{s}{\beta}} r_0 \right)} \right\} + \mathcal{L}^{-1} \left\{ \frac{T_i}{s} \right\} = (T_0 - T_i) \mathcal{L}^{-1} \left\{ \frac{P(s)}{Q(s)} \right\} + T_i \quad (22)$$

where $P(s) = I_0 \left(\sqrt{\frac{s}{\beta}} r \right)$ and $Q(s) = s I_0 \left(\sqrt{\frac{s}{\beta}} r_0 \right)$. Since I_0 is defined as

$$I_0(x) = \sum_{n=0}^{\infty} \frac{\left(\frac{1}{2}x\right)^{2n}}{n! \Gamma(n+1)} \quad (23)$$

it can be concluded both $P(s)$ and $Q(s)$ are polynomial of s . Furthermore, the order of $Q(s)$ is higher than the order of $P(s)$ and the roots of $Q(s) = 0$ are non-repeating,

$$Q(s) = 0 \implies s_0 = 0 \text{ or } I_0 \left(\sqrt{\frac{s_n}{\beta}} r_0 \right) = 0 \quad (24)$$

$$\implies I_0 \left(\sqrt{\frac{s_n}{\beta}} r_0 \right) = J_0 \left(-i \sqrt{\frac{s_n}{\beta}} r_0 \right) = 0 \quad (25)$$

$$\implies J_0(\gamma_n) = 0 \text{ where } \gamma_n = -i \sqrt{\frac{s_n}{\beta}} r_0 \quad (26)$$

$$\implies s_n = -\frac{\beta \gamma_n^2}{r_0^2} \quad (27)$$

Here, γ_n is the n -th root of $J_0(x) = 0$. Evaluate $Q'(s)$ as

$$\frac{d}{ds} \left[s I_0 \left(\sqrt{\frac{s}{\beta}} r_0 \right) \right] = I_0 \left(\sqrt{\frac{s}{\beta}} r_0 \right) + \frac{r_0}{2} \sqrt{\frac{s}{\beta}} I_1 \left(\sqrt{\frac{s}{\beta}} r_0 \right) \quad (28)$$

and the Heaviside expansion of $\mathcal{L}^{-1} \left\{ \frac{P(s)}{Q(s)} \right\}$ is

$$\mathcal{L}^{-1} \left\{ \frac{I_0 \left(\sqrt{\frac{s}{\beta}} r \right)}{s I_0 \left(\sqrt{\frac{s}{\beta}} r_0 \right)} \right\} = 1 + \sum_{n=1}^{\infty} \frac{I_0 \left(\sqrt{\frac{s_n}{\beta}} r \right)}{I_0 \left(\sqrt{\frac{s_n}{\beta}} r_0 \right) + \frac{r_0}{2} \sqrt{\frac{s_n}{\beta}} I_1 \left(\sqrt{\frac{s_n}{\beta}} r_0 \right)} e^{s_n t} \quad (29)$$

$$= 1 + \sum_{n=1}^{\infty} \frac{I_0 \left(\sqrt{\frac{s_n}{\beta}} r \right)}{\frac{r_0}{2} \sqrt{\frac{s_n}{\beta}} I_1 \left(\sqrt{\frac{s_n}{\beta}} r_0 \right)} e^{-\frac{\beta \gamma_n^2}{r_0^2} t} \quad (30)$$

$$= 1 - 2 \sum_{n=1}^{\infty} \frac{J_0 \left(\frac{\gamma_n r}{r_0} \right)}{\gamma_n J_1(\gamma_n)} e^{-\frac{\beta \gamma_n^2}{r_0^2} t} \quad (31)$$

substitute Equation (31) into Equation (22) to obtain

$$T(r, t) = (T_0 - T_i) \left\{ 1 - 2 \sum_{n=1}^{\infty} \frac{J_0 \left(\frac{\gamma_n r}{r_0} \right)}{\gamma_n J_1(\gamma_n)} e^{-\frac{\beta \gamma_n^2}{r_0^2} t} \right\} + T_i \quad (32)$$

$$\frac{T(r, t) - T_i}{T_0 - T_i} = 1 - 2 \sum_{n=1}^{\infty} \frac{J_0 \left(\frac{\gamma_n r}{r_0} \right)}{\gamma_n J_1(\gamma_n)} e^{-\frac{\beta \gamma_n^2}{r_0^2} t} \quad (33)$$